

Part A – Scientific proposal**A.1 Basic details****A.1.1 Title**

6G-INS: Intelligent Network Services for the 6G Era

A.1.2 Abstract (suited for scientific peers, to be uploaded in ISAAC)

The upcoming 6G networks demand unprecedented levels of automation and intelligence to optimize resources, security guarantees, and handling extreme traffic loads. Emerging applications like holographic interactions and digital twins require ultra-reliable, high-throughput networks reaching up to 1 Tbps. 6G-INS addresses these challenges by integrating federated learning (FL) and federated unlearning (FUL) with programmable data planes to offer network service that enables real-time, privacy-preserving AI adaptation. It proposes developing self-regulating mechanisms for dynamic service composition, AI-driven resource optimization, and cross-domain orchestration. By embedding decentralized intelligence into the network fabric, 6G-INS ensures scalable, efficient, and secure service management for next-generation networks.

A.1.3 Summary (suited for non-specialists in multidisciplinary scientific assessment committees)

The rapid expansion of digital applications, such as holographic interactions and digital twins, requires the next generation of communication networks (6G) to handle extreme traffic loads while ensuring security, privacy, and efficiency. To meet these demands, 6G-INS explores new AI-driven approaches to optimize network performance. By integrating federated learning (FL) and federated unlearning (FUL) with programmable network technologies, the project enables secure, adaptive, and efficient service management. Through self-regulating mechanisms and decentralized intelligence, 6G-INS aims to provide a scalable and robust network framework that supports the evolving needs of future digital applications.

A.1.4 Keywords

6G, AI-driven network optimization, network programmability, network service, Federated (Un)learning

A.2 Scientific proposal

A.2.1 Research topic

The fifth generation of communication networks (5G) marked a transformative shift towards softwarization and network function virtualization (NFV), enabling the creation of softwarized virtual networks. This paradigm has facilitated diverse applications by introducing dynamic network slicing and the ability to host multiple virtual networks on a shared physical infrastructure [1-4]. While 5G has driven a profound digital transformation across industry and society, it has also exposed critical challenges in managing the exponentially growing data traffic and ensuring seamless connectivity between user devices, Internet of Things (IoT) systems, vehicular networks, and the edge/cloud continuum [14].

Diverging from its predecessors, the complexity of 6G networks, especially the core network and Radio Access Network (RAN), significantly increases. 6G networks must handle massive connectivity, ensure seamless service delivery, and address key challenges to meet ambitious goals such as ultra-reliable low-latency communications (URLLC), end-to-end quality of service (QoS), scalability for massive connectivity and autonomous systems, and seamless integration of cloud, edge, and fog computing resources. The 6G ecosystem would therefore require unprecedented levels of intelligent and automated management to optimize resources, adapt dynamically to application needs, and handle unprecedented traffic while ensuring data privacy, security, and energy efficiency. For instance, as 6G moves towards the era of "connected intelligence," new applications such as holographic interactions and digital twins demand extreme reliability and throughput (up to 1 Tera bit per second) [13].

These requirements give rise to the following critical challenges:

Challenge 1. End-to-end Quality of Service (QoS) guarantees: The stringent and diverse QoS requirements in 6G networks—such as ultra-low latency, extreme reliability, low-energy and high throughput—pose significant challenges, i.e., ensuring consistent adherence to Service Level Agreements (SLAs) across heterogeneous network slices, applications, and services in a dynamic environment is non-trivial.

Challenge 2. Sustainability in 6G Networks: Sustainability is a growing concern in 6G networks due to the RAN densification (RAN accounts for >85% of the 5G power consumption) and the increasing computational and traffic loads. Demand-aware proactive resource management is fundamental to use network resources only when strictly needed. As device density increases, it is essential to implement energy-saving strategies that reduce the carbon footprint of the network while maintaining high performance and meeting the end-to-end QoS requirements.

Challenge 3. Network service automation and intelligence. Traditional manual network management approaches are no longer viable for the complex and dynamic 6G ecosystem. The integration of AI/ML-driven automation therefore is essential to dynamically configure, monitor, and optimize network services. However, developing lightweight, adaptive AI models for real-time decision-making, especially in resource-constrained environments, presents a major hurdle. While 6G core AI deployment offers a global vision and potentially optimal solution, moving AI to the edge offers lower latency and alleviates the data congestion needed when centralizing the decision-making.

Challenge 4. Data privacy and security in a massively connected landscape: Threat detection as well as data privacy and integrity are considered the most critical issues in 6G networks. Protecting sensitive data from unauthorized access while maintaining network performance requires robust, adaptive security mechanisms. However, achieving this level of security without affecting service speed or efficiency is a major challenge, especially given the sensitivity of data handled in applications like autonomous transportation and healthcare.

By rethinking network service architectures and integrating intelligent, autonomous, and federated algorithms, 6G-INS aims to ensure that next-generation 6G networks can meet the stringent demands of ultra-low latency, high reliability, security, and sustainability. In this proposal, a network service refers to a set of programmable, intelligent, and adaptive mechanisms that manage traffic, connectivity, and compute resources across the 6G infrastructure. These services ensure that diverse applications can operate efficiently while meeting stringent performance and security requirements.

Our primary research question is as follows:

What are the fundamental mechanisms for developing energy-efficient and intelligent network services that effectively leverage heterogeneous resources in 6G networks, while balancing sustainability goals with diverse QoS requirements?

Vision and Main Objectives. The main objective of 6G-INS is to develop a layered and intelligently orchestrated 6G network services that guarantees high security, reliability, and sustainability while fulfilling the stringent QoS requirements of diverse applications. The project focuses on designing advanced network services for both the core network and RAN, leveraging the power of Federated Learning (FL) and Federated Unlearning (FUL) to uniquely realize a real-time AI-driven optimization and orchestration of network programmability. To achieve this vision, 6G-INS breaks down its main objective into specific, well-defined goals that address key technical and operational challenges in 6G network design and management. Our objectives are therefore:

1. to develop an advanced 6G network architecture by designing a modular and flexible network service architecture, focusing on network management, service orchestration, and virtualization. This architecture will prioritize efficient resource distribution and traffic management across diverse applications, meeting high-performance requirements such as URLLC. 6G-INS aims to improve adaptability and responsiveness of network services to dynamically adjust to evolving network conditions and QoS demands.

2. to achieve sustainability via intelligent resource orchestration by developing energy-saving strategies involving multiple RAN features (e.g., resource block allocation, power control, beam management, and access point switch-off). Depending on the variability of the traffic, local and near-time data collection can serve to activate power saving at the edge level, while leveraging centralized data collection at the core provides a more global vision, possibly more suitable to deal with power-saving measures related to user mobility and traffic steering. AI applications, such as Federated Unlearning (FUL) [15], will play an important role in supporting adaptive responses.

3. to enable seamless cross-layer integration for end-to-end 6G services by developing an integrated framework to ensure smooth interaction between upper and lower layers, optimizing resource management and data flow across the protocol stack. This objective aims to bridge the physical-, data link-, and network-layers through cross-layer communication and adaptive resource allocation. The framework will support end-to-end services with high performance and reliability, aligning with the demands of a diverse set of applications, from low-latency industrial automation to high-data-rate media streaming.

4. to strengthen 6G network security with adaptive threat mitigation by developing robust, adaptive security mechanisms across the data link and application layers to safeguard data integrity and confidentiality within 6G networks. This objective focuses on proactive threat detection and response, utilizing AI-driven anomaly detection to identify unusual patterns and potential attacks in real-time. Advanced threat mitigation techniques will enable the system to adapt to evolving security challenges, providing a dynamic response to threats without compromising network performance.

Background and state-of-the-art

To handle vertical-specific demands over the same network infrastructure, the traditional network architecture has been experiencing a softwarization process, moving from the conventional hardware-based infrastructure towards a hybrid hardware-software partitioning of network functionalities [16]. In this direction, the system control via software acts as an enabler towards advanced reconfigurability and flexibility. More specifically, softwarization across all communication layers facilitates the wireless system reconfiguration to accommodate dynamically changing demands, optimize cost and processes, and enable self-management, bringing added value to the wireless infrastructure. However, the rapid proliferation of multi-radio access technology and smart devices has resulted in complex heterogeneous multi-hop mesh network topologies, thus making this reconfiguration and management extremely cumbersome, requiring regular self-optimization of network parameters in response to observed network performance and radio conditions. In parallel, sustainable telecom has been put at the forefront of the EU agenda, considering that telecom infrastructure and technology account for ~4% of the world's electricity consumption [17] and is expected to continue rising with the full deployment of 5G.

Early studies in learning-based techniques have proved to be successful when applied to network optimization and have since then triggered the attention of relevant industry. Federated Learning (FL) [11,19,27,29] is pivotal in realizing a vision where data privacy is paramount, and communication overhead is minimized. By facilitating model training across decentralized devices without centralizing sensitive data, FL is well-suited to the evolving architecture of 6G networks, which are characterized by their integration of diverse and heterogeneous data sources under stringent privacy constraints. FL can be implemented in various ways, such as centralized FL, which relies on a central server for aggregation; distributed FL (DFL), where devices collaborate directly without a central authority; and hierarchical FL (HFL), which uses intermediate aggregators (edge servers) to reduce communication overhead and improve scalability in large-scale systems. These techniques cater to different use cases and network demands [12,22].

Advancing beyond traditional FL, Federated Unlearning (FUL) [15] offers a novel enhancement by enabling the selective removal of model parameters that are influenced by outdated, erroneous, or sensitive data, as per changing data sharing agreements. This capability would be uniquely suited for network operators who require the flexibility to update and refine models continuously without the substantial costs and delays associated with starting afresh. Such an approach is cost-effective; it is essential in dynamic environments like cellular networks where data patterns frequently shift, necessitating agile model adjustments to maintain both operational privacy and security.

AI-native 6G network orchestration. The 3GPP AI-native developments for 6G emphasize the integration of AI-driven techniques within both the near-real-time RAN Intelligent Controller (Near-RT RIC) and the non-real-time RAN Intelligent Controller (Non-RT RIC) [25]. These controllers facilitate distributed data analytics, reducing the need for massive data transfers and enhancing network security [19]. Unlike the current projects, e.g., [20,21,28], or works [22-24,26], 6G-INS will leverage Near-RT RIC and Non-RT RIC advancements to develop AI-based optimization mechanisms, enabling real-time adaptation, enhanced privacy, and efficient resource allocation in future 6G networks.

Scientific originality. The 6G-INS project introduces a novel approach to achieving "connected intelligence" in 6G networks, moving beyond traditional methods by introducing an innovative programmable framework that combines FL, FUL, and network programmability. In typical FL, collaborative training occurs across decentralized nodes through local model updates. The 6G-INS project builds on this concept by embedding programmability into the orchestration of FL and FUL. This allows for the dynamic integration and removal of data points during the learning process. This capability will crucially enable fine-grained control over intelligent network services through programmable data pipelines and decision making, while at the same time enhancing privacy preservation and adaptability.

The **scientific novelty** of 6G-INS lies in the creation of adaptive, flexible, and self-regulating learning algorithms that function seamlessly under the extreme conditions of 6G environments—such as wireless channel variability and resource-constrained mobile edge devices. Unlike existing FL models that rely on centralized aggregation [19,22-24], the innovative combination of FUL with programmable in-network computing enables decentralized unlearning, reducing reliance on a central server and allowing AI-driven adaptation at the network edge.

Moreover, 6G-INS integrates network programmability with AI-driven decision-making, leveraging programmable data planes and in-network computing to optimize performance while reducing energy consumption. By exploiting distributed AI models, and low-power edge devices, the project enhances real-time, energy-efficient, and context-aware decision-making. Low-power edge devices in 6G-INS enhance distributed AI by enabling real-time, energy-efficient decision-making closer to data sources, reducing latency and bandwidth use. By integrating FL and FUL, these devices collaboratively train and refine AI models while maintaining privacy and minimizing communication overhead. Their synergy with programmable data planes and in-network computing ensures adaptive, scalable, and resource-efficient AI-driven network services, aligning with 6G-INS's goal of intelligent, sustainable, and resilient network optimization.

This novel integration of FL, FUL, and programmable networking establishes a transformative framework for adaptive, resource-efficient, and self-optimizing network services. The programmability aspect ensures that 6G-INS is not only capable of real-time intelligence but also offers dynamic reconfiguration, policy-driven control, and automated optimization of network functions. By redefining how FL interacts with programmable infrastructures, 6G-INS positions itself as a key enabler of scalable, resilient, and sustainable 6G services.

A.2.2 Approach

The **6G-INS project** will develop a next-generation intelligent and programmable service management framework for 6G networks, integrating AI-driven adaptability, modular service design, and cross-domain interoperability. Our research is structured into four interdependent technical work packages (WPs), each addressing a key technical pillar that collectively drives the project's overarching goal: delivering efficient, secure, and scalable network services.

WP1 establishes the foundational system design, defining the interplay between core and RAN services, AI-driven programmability, and cross-layer intelligence. This ensures a flexible, modular, and future-proof framework that supports diverse 6G applications. As shown in **Figure 1**, the development of 6G-INS's network services will follow a modular approach to maximize efficiency and effectiveness. This modular design offers several advantages: (1) Each network-layer service will be tailored to address specific system challenges identified during initial analysis. (2) Compatibility between various services will be enhanced, facilitating smoother interactions and reducing integration complexity. (3) The framework

will be designed for seamless integration with existing 6G network technologies, ensuring backward compatibility and adaptability. This structured methodology ensures that individual network services are developed and tested in isolation for functional accuracy before being incrementally integrated into a cohesive system. This phased integration enables continuous validation of each module, allowing adjustments to be made as needed. This approach ensures that the final system is not only robust and secure but also capable of meeting the stringent QoS requirements of advanced 6G applications. The architecture of 6G-INS is designed around two subsets of network services: (1) RAN network services and (2) core network services. Each subset is tailored to achieve specific objectives crucial to the 6G-INS framework.

WP2 develops federated AI mechanisms that optimize resource allocation, enhance network automation, and enable dynamic adaptation to changing service demands. This WP directly feeds WP3 and WP4, providing intelligent decision-making capabilities.

WP3 focuses on designing and implementing modular and QoS-aware network functions that seamlessly integrate into the 6G-INS architecture. These services will be aligned with the AI-based orchestration from WP2.

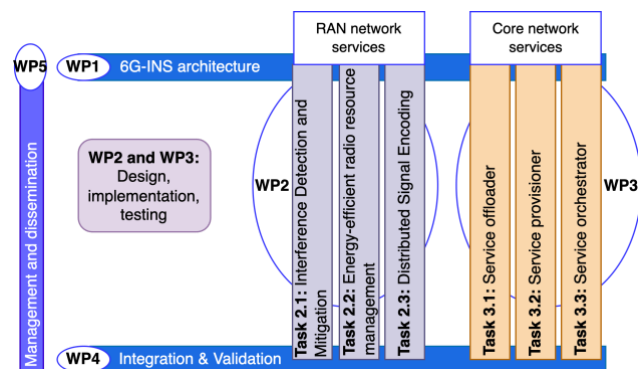


Figure 1 The WP structure of 6G-INS

WP4 ensures the modular system's compatibility and adherence to 6G's stringent latency, security, and scalability demands. By leveraging a dynamic feedback loop with other work packages, WP4 enables continuous optimization, real-time adaptation, and intelligent self-regulation, fostering an autonomous and highly efficient network ecosystem.

This modular and AI-driven approach ensures that each WP directly contributes to the overall objective—creating an adaptive, intelligent, and programmable 6G service framework. Instead of isolated developments, each component is designed for continuous interaction, reinforcing a cohesive ecosystem where programmability, intelligence, and interoperability converge to meet 6G's stringent requirements.

We now describe each WP in more detail in Table 1.

Table 1 Description of WPs		
WP number: 1	WP title: 6G-INS architecture	Duration: M1-M48
Objectives: (1) to develop a modular architecture that integrates the physical, data link, and network layers to support resource management, edge computing, and IoT compatibility. This ensures a programmable and adaptable foundation for 6G-INS, supporting AI-driven network automation and dynamic service provisioning. WP1 serves as the architectural backbone, ensuring seamless coordination between RAN (WP2) and core network services (WP3). (2) to define modular network services adaptable to diverse 6G use cases, facilitating URLLC. These services will leverage programmability and AI to dynamically optimize performance, ensuring interoperability across RAN (WP2) and core networks (WP3). Description of Work. Both partners will participate in this WP, which will be executed by PhD1 at Eindhoven University of Technology (TU/e) and PhD2 at the University of Luxembourg (Uni.lu), who will jointly define the architecture to ensure seamless cross-layer functionality, with each institution focusing on optimizing specific network services. Task 1.1 6G-INS system architecture. This task involves creating a high network service architecture based on the requirements of application focusing on latency, reliability, and security needs. Task 1.2 Validation of architecture. This task involves reviewing and validating requirements and architecture to ensure alignment with project objectives. This ensures alignment with design objectives before detailed implementation. The outcome will be comprehensive documentation detailing requirements, analysis, and system architecture.		
WP number: 2	WP title: RAN network services	Duration: M3-M36
Objectives: (1) to develop robust solutions that integrate the physical- and data link-layers to support advanced resource management, edge computing, and IoT compatibility, while embedding the innovative FL/FUL framework to enable dynamic, privacy-preserving data processing at the network edge. (2) to enhance signal processing techniques and error correction methods to optimize data transmission throughput and reliability in 6G networks, leveraging adaptive federated learning and unlearning methods to adjust to changing network conditions in real-time. PhD2 at Uni.lu will carry out the tasks of this WP. Description of Work. WP2 targets the development of robust network protocols and algorithms that operate at the lower network layers of the 6G networks. This work package explicitly incorporates the project's scientific originality by integrating FL and FUL mechanisms into the layered design and operation of RAN, resource allocation, and distributed encoding strategies. These techniques are adapted to function under the dynamic and challenging conditions characteristic of 6G environments, such as wireless channels and limited computational resources at mobile edge devices. Task 2.1 Threat and anomaly data detection & mitigation: This task focuses on developing real-time threat detection and mitigation attacks capable of adapting to dynamic spectrum environment of 6G networks. ML models such as convolution neural network (CNNs), will be used to analyze and classify threat patterns. In parallel, the development of secured-data flow via FL will enable collaborative improvements in anomaly data detection across distributed nodes, while the FUL mechanism will dynamically remove outdated and irrelevant data without re-training from scratch. This combined approach not only enhances the system's adaptability but also maintains data privacy throughout the network. Task 2.2 Energy-efficient learning-driven RAN management: To reduce energy consumption and improve reaction times to network events, this task is dedicated to exploring Near-RT RIC and Non-RT RIC RAN orchestration, with layered AI engines to govern RAN features such as power, radio-block assignment and antenna adaptability [19]. Reinforcement Learning (RL) models, developed using OpenAI Gym combined with Pytorch/Tensorflow, will be used to predict and adapt to fluctuation in network traffic. The use of FL enables decentralized learning across edge devices, while FUL facilitates the unlearning of obsolete patterns as network conditions evolve. Together, these mechanisms achieve a resilient and energy-efficient radio resource management system that can dynamically respond to the real-time demands of 6G networks.		

Task 2.3 Distributed signal encoding: In this task, the focus is on enhancing the robustness and efficiency of data transmission through the development of distributed signal encoding strategies that operate in association with FL/FUL. The goal is to implement encoding techniques that can dynamically adjust to the network's conditions, thus reducing redundancy and improving throughput. By leveraging the collaborative capabilities of FL, distributed nodes will optimize encoding parameters in real-time, while FUL will ensure that any outdated or malicious data contributions are directly removed. This integrated approach facilitates a highly adaptable encoding process that is well-suited to the complex and rapidly changing environment typical of 6G networks. The performance of these approaches will be tested on a using MATLAB /Python) and in TeleComAI-Lab [18].		
WP number: 3	WP title: Core network services	Duration: M3-M36
Objectives: (1) to build network services to meet the QoS requirements of 6G applications. (2) to implement the network services on programmable switches that can handle a massive number of users with a huge amount network traffic. (3) to verify and validate the implemented functionality of each network service. PhD1 at TU/e will carry out the tasks of this WP.		
Description of work: This WP focuses on designing, implementing, and testing network services that meet the functionality requirements of 6G network applications. Task 3.1. Resource allocator. This task will develop methods to efficiently allocate and manage limited data-plane resources within programmable network devices to various services while ensuring accurate measurement and monitoring of network slices. We will leverage compact data structures within P4-enabled switches to track network slice-specific telemetry data while minimizing memory and computational overhead. To maximize the accuracy of slice-level resource allocation decisions, we will analyze the trade-offs between measurement precision and data-plane resource usage, ensuring that key network performance indicators remain reliable. When data-plane resources cannot accommodate all monitoring queries, we will design hybrid approaches that partition queries between the data plane and control-plane software. Additionally, we will explore efficient aging mechanisms to handle long-running queries, ensuring that stale measurement data expires or decays over time without consuming excessive resources. We will implement the resource allocator on P4 and extend Berkeley Packet Filter (eBPF) to have the flexibility to run it on the network switches and end-hosts. Task 3.2 Service orchestrator. This task will develop a network-wide monitoring framework for 6G networks, enabling real-time detection of large-scale anomalies such as DDoS attacks and congestion. Instead of isolated measurements of traffic patterns per network devices, we will implement a one-big-switch abstraction, aggregating telemetry from multiple programmable devices to provide a holistic view of network conditions based on the results of task 3.1. We will design a resource-aware and traffic-aware service placement strategy based on FUL, dynamically distributing monitoring tasks across switches and hosts. Instead of fixed-frequency telemetry reporting, we will develop an AI-driven probabilistic reporting mechanism that adjusts reporting frequency based on anomaly likelihood and network conditions. Measurement data will be collected using P4-based telemetry on Tofino switches and eBPF-based host monitoring, aggregated via gRPC streaming to an SDN controller. Task 3.3 Self-healing service. This task will focus on developing an AI-driven, self-healing mechanism for 6G networks, allowing the network to autonomously detect root cause of QoS degradation and implement recovery actions without manual intervention. We will develop advanced and FUL-based inference algorithms to find out the root cause of issues occurring within programmable network devices based on the collected data from compact data structures, using P4 to enable fine-grained control and monitoring of network behavior. Once root causes are detected, AI-based decision-making algorithms will initiate corrective actions, such as rerouting traffic, scaling resources, or switching to backup network paths. To enable self-healing, we will utilize the control plane of the programmable networks to configure and adapt network infrastructure. AI models for fault detection and recovery decisions will be developed using TensorFlow or PyTorch, and the system will integrate with existing network monitoring platforms like Grafana for visualization and alerting. We will test the functionality of the service in a network testbed consisting of high-performance servers connected to Tofino programmable switches.		
WP number: 4	WP title: Integration and validation	Duration: M12-M45
Objectives: (1) to integrate all services into the 6G-INS system based on the 6G-INS architecture in WP1 to facilitate seamless operation across physical, data link, and network layers. (2) to enable smooth interaction and secure data flow between all components, preparing the system for testing and validation. (3) to validate the system's performance, reliability, and security features to ensure it meets the objectives for 6G network services.		
Description of work: Both PhD students will collaborate on system integration, ensuring each component's cross-layer interoperability. The network software/hardware testbed from TU/e will be considered as baseline for core network services and the RAN network solutions implemented by Uni.lu. Task 4.1: Service integration. This task aims to ensure seamless integration between innovations at the physical and data link layers with network layer services. An integration framework will be developed to define data flow and		

interaction points across these layers. Finally, integration tests will be conducted in controlled environments to evaluate the end-to-end QoS and ensure alignment with overall system objectives.

Task 4.2: Interface specification and validation. Drawing on the integration architecture from Task 4.1, this task will focus on specifying and refining the definitions of various interfaces. It will also include validation to ensure that these interfaces meet the required standards and facilitate seamless interaction among services.

Task 4.3 Validation of RAN network services. This task involves designing and executing testing methods for the validation of the network services related to RAN network services. The focus will be on assessing both real-time performance and accuracy.

Task 4.4 Validation of core network services. This task entails designing and executing testing methods and systems for the validation of core network service associated with 6G-INS. Particular emphasis will be placed on scaling and efficiency of network services.

Task 4.5 Refine architecture. This task will perform refining the high-level architecture based on technical contributions, incorporating insights from the protocol stack layer optimizations.

WP number: 5	WP title: Project management and dissemination	M1-M48
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Objectives: (1) to provide the overall project administration and management support. (2) to monitor the R&D progress and activities to ensure adherence to the overall project plan. (3) to provide coordination of project reporting activities.

Description: This WP comprises the overall project management, planning, coordination, and communication with the NWO and FNR, as well as quality monitoring of the research program.

Task 6.1 Project management: This task will provide the overall project coordination and ensure the day-to-day management, including activities such as preparation and support to the project meetings; resource monitoring against project deliveries and results; coordination of reporting and administrative activities, etc.

Task 6.2 Project dissemination: We will disseminate research findings through academic publications, conference presentations and technical reports. A measure of the success of the scientific excellence of the project will be confirmed by the number and quality of the papers published jointly by the partners and their contributions and/or invited lectures presented at leading international conferences. An interactive and frequently updated website will be used as the primary dissemination channel to the public. Project news will also be announced through dedicated social media accounts. We will reach out and engage with the public to create awareness of 6G-INS. Two PhD will be graduated within this proposal and their theses will be publicly available.

Task 6.3: PhD theses. Both PhD students will finalize their PhD theses based on the results obtained during the project.

The 6G-INS project will run for four years, and Figure 2 shows Gantt chart and the work plan for each WP and its tasks.

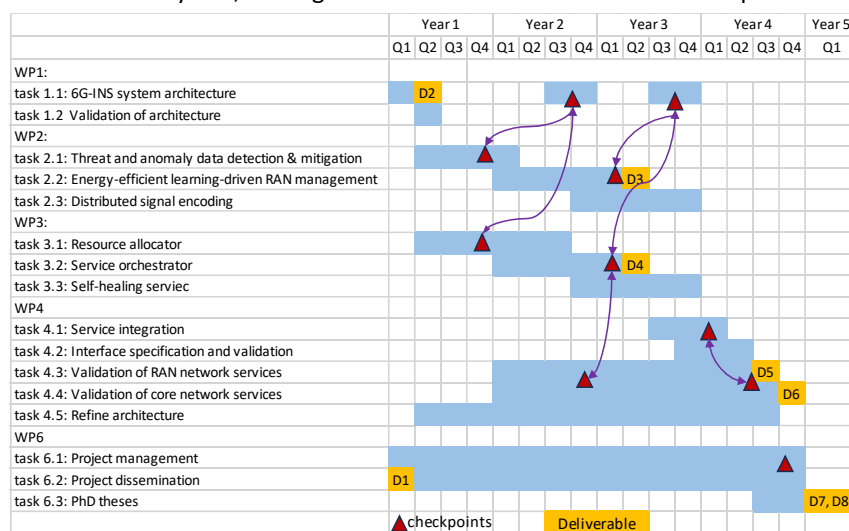


Figure 2 Gantt chart of 6G-INS

Running of the above tasks in different WPs will result in the following deliverables.

Table 2 Deliverables List					
No.	Deliverable Title	Short Description	WP	Lead	Due Date
D1	6G-INS website	6G-INS website	1	TU/e	M3
D2	The architecture of 6G-INS	The architectural description of 6G-INS	2	TU/e	M18
D3	RAN network services	RAN network services	2	Uni.lu	M25

D4	Core network services	Core network services	3	TU/e	M34
D5	Validation of RAN network services	Validation of RAN network services	2	Uni.lu	M42
D6	Validation of core network services	Validation of core network services		TU/e	M45
D7	PhD thesis 1	PhD thesis 1	5	TU/e	M52
D8	PhD thesis 2	PhD thesis 2	5	Uni.lu	M52

A.2.3 Justification

Personnel. We hire two PhD students—one at TU/e and one at Uni.lu—to work on complementary aspects of 6G network services as the project requires long-term foundational research, methodological development, and expertise-building in AI-driven 6G programmability. The PhD at Uni.lu will focus on programmable network services of RAN, including in-network computing and federated unlearning in the data plane. The PhD at TU/e will work on core network services, such as adaptive service orchestration. Both PhD students will collaborate through joint research meetings, cross-institutional visits, and shared experimentation on testbeds. Knowledge exchange will be facilitated through co-authored papers and integration of their methodologies into a unified framework. Each PhD will have a dedicated supervisors at their home institution, with regular joint supervision meetings involving both teams to ensure alignment. TU/e and Uni.lu already have a collaboration structure in place to support mobility and knowledge sharing.

Knowledge utilization. The project emphasizes making research outputs broadly available to the scientific and industrial community, ensuring practical and theoretical impact. The allocated budget will include costs for open-access publication of results in high-impact journals and conference proceedings. This approach ensures wide dissemination of findings, fostering collaboration, innovation, and practical application in related fields. Additionally, the budget will support the development of publicly available datasets and open-source software tools to maximize the reach and utility of the project outcomes.

Internationalization. The budget includes provisions for fostering international collaboration between the TU/e and the Uni.lu, ensuring a seamless exchange of knowledge, ideas, and resources. This cost includes travel and accommodation expenses for scientific meetings, workshops, and joint development sessions, as well as mobility exchanges between the two institutions. Specifically, we allocate budget for three-month exchanges (Uni.lu → TU/e, TU/e → Uni.lu), facilitating in-depth collaboration and strengthening the impact of the project outcomes on an international scale. These collaborative activities are essential to maintaining alignment between the partners, leveraging shared expertise, and enhancing the overall success of the project.

Small investment. A small investment is planned to procure a high-performance computing server capable of generating high-speed network traffic and network switches. This server will be used to test the functionality of the developed network services under high-traffic workloads and is integral to the project's experimental validation phase. While two servers are currently available at the IRIS lab of TU/e, they are predominantly allocated to other projects, limiting their availability for this initiative. The planned procurement aligns with the project's objectives to simulate real-world 6G scenarios and evaluate the scalability and robustness of the proposed solutions. Additionally, this investment can support other research within the IRIS cluster when the servers are not in use for this project.

A.2.4 Embedding of research and relevant expertise of researchers involved

Expertise

Dr. Mostafaei, the Principal Investigator at TU/e, is a leading expert in networked systems and programmability. He has specialized training in Intel's Tofino architecture, which he has successfully leveraged across multiple projects and collaborations to translate cutting-edge hardware programmability into practical, high-performance network solutions [30,31]. During his PhD, he developed several innovative network architectures and frameworks to provide advanced network services for Internet Service Providers and Internet Exchange Points. Dr. Mostafaei has extensive experience in mentoring graduate students, having supervised numerous successful projects. He will co-supervise the PhD candidate at TU/e alongside Prof. N. Meratnia, who will also serve as the promotor, ensuring strong interdisciplinary guidance and alignment with the project's scientific and technological objectives.

Dr. Lagunas, the Principal Investigator at Uni.Lu, is an expert on demand-aware interference-limited wireless network optimization, with a particular focus on non-convex optimization and artificial intelligence assisted optimizers. She has been leading the AI-related research line in SIGCOM group, with the creation of the TelecomAi-Lab. She actively participates as TPC member of relevant conferences (e.g. ICC, GlobeCom) and organizes Workshops related to AI technology (ICC'25). She is Publication Chair of the IEEE International Conference on Machine Learning for Communication and Networking (ICMLCN'25); "Wireless, Optical and Satellite Networks (WOS)" track chair at EUCNC - 6G Summit 2025, and member of the best student paper award committee. She is founding member of a new Special Interest Group (SIG) in IEEE Communications Society on AI-Driven Integration of Terrestrial and Non-Terrestrial Networks. Finally, she is member of the EURASIP Board of Directors since 2025. Dr. Lagunas will jointly co-supervise the PhD student of Uni.Lu with Dr. F. Ortiz and Dr. O. Aouedi.

Collaborators

The success of the 6G-INS project relies on the collaboration between our teams at TU/e and Uni.lu. This partnership is based on our common interests in advancing network service management and AI-driven automation, and on a history of effective collaboration and complementary expertise in critical areas relevant to the 6G landscape.

TU/e brings extensive experience in networked systems, programmability, application of AI in wireless networking. This technical expertise is vital for the development of advanced network architectures and the integration of AI/ML driven automation into the 6G network services we intend to design, develop, and test.

Uni.lu brings expertise in Federated Learning (FL) and AI in Non-Terrestrial Networks, with significant prior research on the optimization of satellite communication systems using machine learning techniques. This expertise is crucial for developing the intelligent network services that are at the core of the 6G-INS project.

Through this collaborative framework, the 6G-INS project is poised to integrate and leverage the distinct but complementary strengths of each partner, driving innovation and ensuring the successful implementation of the proposed research activities.

The 6G-INS project prioritizes strong collaboration between partners to ensure successful outcomes. To this end, we will implement a PhD exchange program to promote cross-institutional learning and joint research development. Workshops will be held annually at both Eindhoven and Luxembourg, serving as platforms for knowledge sharing, research alignment, and community engagement. Additionally, we plan to co-organize special sessions at leading international conferences to disseminate project results and foster interdisciplinary collaboration.

Infrastructure

The Interconnected Resource-aware Intelligent Systems (IRIS) cluster at the Department of Mathematics and Computer Science of TU/e owns high-performance cluster servers and programmable network switches that will be used to test and verify the performance and operation of 6G-INS. Each programmable Tofino switch has the capacity up to 3.2 Tbps and will be used for 6G-INS validation.

The University of Luxembourg (Uni.lu) operates since 2007 a large academic HPC facility that remains one of the reference HPC implementation within the country, offering a cutting-edge research infrastructure to Luxembourg public research while serving as edge access to the Euro-HPC Luxembourg supercomputer operated by LuxProvide and more focused at serving the private sector. A special focus was laid for the ULHPC facility on the development of large computing power combined with huge data storage capacity to accelerate the research performed in intensive computing and large-scale data analytics (Big Data).

TelecomAI-Lab of the University of Luxembourg focuses on developing and validating AI-based solutions for telecommunications, particularly in wireless and satellite communication systems. Its research explores neuromorphic computing paradigms for energy-efficient and high-performance signal processing, as well as AI-driven enhancements for 5G/6G networks and non-terrestrial networks (NTN). The lab also investigates AI-based optimizations in physical layer communications, resource allocation, and interference management. In addition, the infrastructure of the Uni.lu, including the high-performance computing facilities and the 6GSpaceLab, provides essential resources for the simulation and testing phases of our project.

A.2.5 Risk assessment

Providing network services in 6G within the 6G-INS context poses several inherent risks. The key risks and their potential mitigations are outlined in the following table.

Type	Risk	Mitigation
Technical Risks		
Network Service Integration	Integrating RAN and core network services may result in incompatibility issues between different protocol layers.	- Define clear interface specifications in WP1 and WP4 for seamless cross-layer service integration. - Conduct incremental validation and testing throughout the development phases.
AI Model Adaptability	The FL and FUL techniques may struggle to adapt to real-time 6G network conditions.	- Implement periodic model fine-tuning using real-time data.
Latency and QoS Guarantees	Ensuring URLLC across 6G core network services is challenging.	- Optimize protocol designs in WP2 for energy-efficient radio resource management. - Utilize AI-based QoS adaptation strategies in WP3 to dynamically allocate resources.
Security & Privacy	Massive IoT connectivity and distributed learning increase vulnerability to cyber threats and data leakage.	- Implement AI-driven anomaly detection in WP3 for proactive threat mitigation.

		- Use differential privacy and secure aggregation techniques in federated learning models.
Non-Technical Risks		
Collaboration & Coordination	Managing research collaboration between TU/e and Uni.lu may face logistical and administrative hurdles.	- Establish structured research meetings and annual workshops (WP5). - Regular monthly meetings between international partners and PhD exchange program for knowledge transfer and team alignment.

A.3 Scientific and/or societal impact

Applicants in the Open Competition Domain Science – M can choose to put primary focus on scientific impact, societal impact, or a combination thereof. This application opts for: Scientific and societal impact of comparable focus.

We opt for both scientific and societal impact because 6G-INS advances the state-of-the-art in network intelligence, programmability, and AI-driven automation, contributing to fundamental research in 6G architectures. Simultaneously, our reliable, low-latency communication, and energy-efficient network services provide direct benefits to society, enabling critical applications such as smart healthcare, autonomous systems, and sustainable digital infrastructure. By bridging cutting-edge research with real-world applications, 6G-INS ensures that next-generation networks are not only technologically advanced but also meet societal needs.

Scientific Impact. The 6G-INS project tackles key challenges in applying distributed AI to 6G networks by developing and integrating advanced FL/FUL and distributed deep learning models. Specifically, the project will design adaptive algorithms that enable FL/FUL to function efficiently in dynamic network topologies with strict latency and resource constraints. By introducing mechanisms for intelligent resource allocation, energy-efficient model training, and secure data aggregation, 6G-INS enhances scalability and privacy in distributed learning environments.

To support these advancements, 6G-INS will prototype and validate its methods in real-world scenarios, using testbeds and simulation environments that replicate 6G conditions. This includes evaluating performance in varying network loads, assessing privacy-preserving capabilities through differential privacy techniques, and optimizing FL/FUL architectures for heterogeneous devices.

The project's impact extends beyond theoretical research by making open-access source code, datasets, and frameworks available to the community. This allows researchers to refine and expand the methodologies introduced, fostering collaboration across disciplines such as edge AI and network programmability. Through this open approach, 6G-INS ensures its contributions are practical, verifiable, and extendable for future 6G systems.

Moreover, 6G-INS will serve as a foundation for new interdisciplinary research initiatives within the participating institutes, fostering collaborations with experts in AI, communication systems, and system optimization. The scientific impact of this project will not only advance the understanding of intelligent network systems but will also shape the trajectory of AI research by introducing benchmarks and methodologies for energy-efficient, privacy-conscious, and resilient AI systems.

Societal Impact. The 6G-INS project has the potential to address critical societal challenges, contributing to sustainability, data privacy, and user-centric network design. By leveraging energy-efficient AI algorithms like FL/FUL, the project directly supports global efforts to enhance sustainability in the ICT sector. Distributed learning techniques eliminate the need for centralized data processing, significantly reducing energy consumption and carbon footprints, while maintaining high levels of performance. This innovation promotes environmental sustainability by making intelligent connectivity not only more efficient but also eco-friendly, aligning with broader sustainability goals in the context of 6G networks.

Incorporating FL and distributed AI models in 6G-INS prioritizes privacy and data security, addressing growing societal concerns about data misuse and breaches. These privacy-aware approaches ensure that sensitive user data remains decentralized, reducing risks while complying with evolving regulations such as GDPR and similar frameworks globally. By promoting trust and transparency in AI-driven solutions, the project advances ethical AI practices and fosters public confidence in next-generation communication systems.

Furthermore, the user-centric approach of 6G-INS contributes to enhancing the quality of life for individuals and communities by ensuring seamless, reliable, and context-aware network services. This translates into tangible societal benefits such as improved access to healthcare, education, and remote work opportunities through intelligent connectivity. By optimizing network services to meet diverse user requirements, 6G-INS enhances inclusivity and accessibility, particularly in underserved regions.

Through its dual focus on scientific innovation and societal well-being, the 6G-INS project exemplifies a transformative approach to addressing the challenges of the future while contributing to sustainable, secure, and equitable technological advancements.

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Part B – Applicants and additional information

B.1 Research areas

Research area 1: Computer systems, architectures, networks 80%

Optional: Research area 2: Artificial intelligence, expert systems 20%

Optional: Research area 3: Choose Research Area Weight %

Optional: Research area 4: Choose Research Area Weight %

Optional: Research area 5: Choose Research Area Weight %

B.2 Main applicant – additional information.

Name (title, initials, first name, last name): **dr. H. (Habib) Mostafaei**
 Affiliation - university/institute: **Eindhoven University of Technology**
 Affiliation - department: **Department of Mathematics and Computer Science**
 Position: **Assistant professor**
 Paid position: **Yes**
 Type of contract: **permanent**
 End date of contract: **-**

Co-applicant – additional information

Name (title, initials, first name, last name): **dr. E. (Eva) Lagunas**
 Affiliation - university/institute: **University of Luxembourg**
 Affiliation - department: **SnT, Interdisciplinary Center for Security, Repliability and Trust.**
 Position: **Other: Research Scientist**
 Paid position: **Yes**
 Type of contract: **permanent**
 End date of contract: **-**

B.3 Other pending or future grant applications on the same research topic

No

B.4 Public summaries (also needs to be uploaded in ISAAC)

Dutch title and summary

6G-INS: intelligente netwerk services voor het 6G tijdperk

De snelle uitbreiding van digitale toepassingen, zoals holografische interacties en digital twins, vereist dat de volgende generatie communicatienetwerken (6G) extreme verkeersbelastingen aankan en tegelijkertijd veiligheid, privacy en efficiëntie waarborgt. Om aan deze eisen te voldoen, onderzoekt 6G-INS nieuwe AI-gestuurde benaderingen om de netwerkprestaties te optimaliseren. Door federated learning (FL) en federated unlearning (FUL) te integreren met programmeerbare netwerktechnologieën, maakt het project veilige, adaptieve en efficiënte servicemanagement mogelijk. Met zelfregulerende mechanismen en gedecentraliseerde intelligentie streeft 6G-INS ernaar een schaalbaar en robuust netwerkraamwerk te bieden dat aansluit bij de evoluerende behoeften van toekomstige digitale toepassingen.

English title and summary

6G-INS: Intelligent Network Services for the 6G Era

The upcoming 6G networks demand unprecedented levels of automation and intelligence to optimize resources, ensure security, and handle extreme traffic loads. Emerging applications like holographic interaction and digital twins require ultra-reliable, high-throughput networks reaching up to 1 Tbps. 6G-INS addresses these challenges by integrating

federated learning (FL) and federated unlearning (FUL) with programmable data planes to offer network service that enables real-time, privacy-preserving AI adaptation. It proposes developing self-regulating mechanisms for dynamic service composition, AI-driven resource optimization, and cross-domain orchestration. By embedding decentralized intelligence into the network fabric, 6G-INS ensures scalable, efficient, and secure service management for next-generation networks.

B.5 Questions Data management section

1. Will this project involve re-using existing research data?

- ☐ Yes: From my own or a collaborator's prior research.
- ☐ Yes: Publicly available data.
- ☒ No: Have you considered re-using existing data but discarded the possibility? Why?

If no, please briefly explain why; if yes, state any constraints on re-use of existing data if there are any.

We will generate network data traffic using software tools like fastclick to test the performance of designed network services under various network traffic conditions. We will make the code of traffic generator publicly available.

2. Will data be collected or generated that are suitable for reuse?

- ☒ Yes: Please answer questions 3 and 4.
- ☐ No: Please explain why the research will not result in reusable data or in data that cannot be stored or data that for other reasons are not relevant for reuse.

3. After the project has been completed, how will the data be stored for the long-term and made available for the use by third parties? Are there possible restrictions to data sharing or embargo reasons? Please state these here.

After the completion of the project, all collected data will be stored in a secure, long-term storage facility, such as the SURF Data Repository and 4TU repository, which adheres to international standards for data preservation and management. The data will be accessible through the repository, accompanied by thorough metadata and documentation to facilitate its reuse by third parties. Within Interconnected Resource-aware Intelligent Systems (IRIS) cluster of Eindhoven University of Technology, we have a research data management (RDM) policy which ensures continuity, reproducibility, and verification of research output. We plan to make the data publicly available under an open-access model, with a possible embargo period of 24 months to accommodate publication processes. Any sensitive personal data will be anonymized and shared in compliance with GDPR. No further restrictions on data sharing are foreseen, except for the aforementioned embargo for scientific publishing.

4. Will any costs (financial and time) related to data management and sharing/preservation be incurred?

- ☐ Yes: Then please be sure to specify the associated expenses as 'implementation cost' within the budget post 'Material costs' in the budget table of this proposal.
- ☒ No: All the necessary resources (financial and time) to store and prepare data for sharing/preservation are or will be available at no extra cost.